

Notes: Haushalter, Klasa, and Maxwell (JFE, 2007)
**The Influence of Product Market Dynamics on a Firm's Cash Holdings and
Hedging Behavior**

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1 Big picture

This paper studies whether firms use financial policies to manage **predation risk**: the risk that, when a firm cannot fund valuable investment opportunities, rivals can invest aggressively, capture market share, and weaken the firm in product markets. The authors focus on two policies that can preserve internal financing capacity: **cash holdings** and **currency derivatives**.

Main finding

The core result is that firms appear to manage predation risk. Firms whose investment opportunities are more interdependent with rivals hold more cash and are more likely to use currency swaps. These policies are not just passive balance-sheet choices; they are linked to investment behavior during industry downturns.

- Across industries, firms in more concentrated industries hold more cash and are more likely to use currency swaps.
- Within industries, firms closer to the technological core of the industry and firms whose stock returns co-move more with industry stock returns hold more cash and hedge more.
- The cash-holdings results are stronger in industries with larger growth opportunities, where predation risk should matter more.
- Cash-rich firms facing high predation risk are more likely to increase investment when industry-level investment falls.
- Cash holdings and derivatives appear to be substitutes in this product-market context: firms with higher industry-adjusted cash are less likely to use currency swaps.

Interpretation: financial policy is part of product-market strategy. Cash and hedging can help firms maintain investment capacity when rivals are vulnerable, or avoid becoming vulnerable when rivals are strong.

2 Incremental contribution of the paper

1. **Links predation risk to corporate financial policy.** Earlier work documents that financially weak firms can be attacked by rivals. This paper asks whether firms anticipate this threat and choose cash and hedging policies accordingly.
2. **Moves beyond industry concentration alone.** The paper measures investment-opportunity interdependence with three proxies: industry concentration, technological similarity, and the correlation between firm and industry stock returns.
3. **Shows both inter-industry and intra-industry evidence.** The paper demonstrates that product-market dynamics matter across industries and among firms within the same industry.
4. **Connects financial policy to real investment behavior.** The paper does not stop at cash and derivatives choices; it shows that cash-rich, high-predation-risk firms are more likely to raise investment when industry investment declines.

5. **Clarifies the relation between cash and derivatives.** The paper provides evidence that, when cash and derivatives serve similar product-market risk-management roles, they can act as substitutes.

3 Research question and hypotheses

The paper asks: *Do firms choose cash holdings and derivatives to protect themselves against product-market predation?*

3.1 Hypothesis 1: predation risk and financial policy

Firms with more interdependent growth opportunities with rivals should be more likely to hold cash or use derivatives. The logic is that such firms face greater consequences if they cannot fund investment when rivals can.

The empirical predictions are:

- firms in more concentrated industries should hold more cash and hedge more;
- firms with operations closer to the industry technological core should hold more cash and hedge more;
- firms whose growth opportunities co-vary more with industry rivals should hold more cash and hedge more.

3.2 Hypothesis 2: cash and derivatives as substitutes

If cash holdings and derivatives both reduce product-market exposure, they can substitute for each other. In particular, if a firm already has high cash holdings relative to rivals, it may have less need to hedge with derivatives.

4 Data and measurement

4.1 Samples

- **Cash holdings sample:** Compustat manufacturing firms from 1993 to 2001.
- **Derivatives sample:** S&P 500 manufacturing firms from 1993 to 1997. The dependent variable is whether the firm uses currency swaps.
- **Industry concentration data:** 1997 U.S. Economic Census manufacturing concentration ratios, measured at the six-digit NAICS level.

4.2 Predation-risk proxies

The authors use three proxies for the interdependence of a firm's investment opportunities with rivals.

- **Industry concentration:** Herfindahl-Hirschman Index and four-firm concentration ratio. Predation risk is expected to be higher in more oligopolistic industries.

- **Technological similarity:** absolute deviation between a firm’s capital-to-labor ratio and the median ratio in its industry, scaled by the industry range. A smaller deviation means the firm is closer to the industry technological core.
- **Firm-industry stock return correlation:** estimated from firm monthly stock returns and industry stock returns. A higher correlation proxies for growth opportunities that co-vary more with rivals.

4.3 Main policy variables

- **Cash holdings:** cash and short-term investments, often scaled by assets or transformed as the log of cash relative to noncash assets.
- **Currency swap usage:** an indicator equal to one if the firm uses currency swaps in a given year.
- **Investment behavior:** whether capital expenditures plus R&D increase when industry-level investment declines.

5 Empirical design

5.1 Cash and derivatives regressions

The main tests relate cash holdings or currency swap usage to the three predation-risk proxies. The cash regressions use OLS specifications, including pooled regressions, time-series averages, and firm fixed effects where appropriate. The currency-swap regressions use probit models.

The specifications control for standard determinants of cash and hedging, including foreign sales, number of business segments, firm size, market-to-book assets, R&D, capital expenditures, leverage, bond ratings, dividend status, profitability, and industry operating-income volatility.

5.2 Growth-opportunity interaction tests

If the interpretation is really predation risk, the relation between interdependence and cash holdings should be stronger when there are more growth opportunities at stake. The authors test this by interacting predation-risk proxies with a high industry market-to-book dummy.

5.3 Investment behavior tests

The paper then studies whether financial policy has strategic benefits. It asks whether firms with high cash holdings and high predation-risk exposure are more likely to increase investment when industry investment falls. This test is meant to capture the idea that cash-rich firms can invest while rivals are forced to cut back.

5.4 Cash versus derivatives tests

Finally, the paper tests whether cash and derivatives are substitutes. It estimates currency-swap usage as a function of industry-adjusted cash holdings and the dispersion of cash holdings among industry rivals.

6 Tables and figures

6.1 Industry concentration and univariate policy patterns

Table 1
Industries sorted by Herfindahl-Hirschman Index values
For each three-digit NAICS industry within the manufacturing sector, we report the median value for the Herfindahl-Hirschman Index for the six-digit NAICS industries within the particular three-digit NAICS industry.

Three-digit NAICS code	Industry name	Median six-digit NAICS industry Herfindahl-Hirschman Index value
323	Printing and related support	77
321	Wood products	193
332	Fabricated metal products	238
337	Furniture and related products	301
313	Textile mills	317
339	Miscellaneous	332
333	Machinery	378
314	Textile product mills	379
326	Plastics and rubber products	396
324	Petroleum and coal products	434
315	Apparel	491
327	Nonmetallic mineral products	562
331	Primary metal	599
334	Computer and electronic products	660
322	Paper	671
311	Food	684
335	Electrical equipment, appliances, and components	765
316	Leather and allied products	849
325	Chemicals	857
312	Beverage and tobacco products	987
336	Transportation equipment	1133

Table 1. Industries sorted by Herfindahl-Hirschman Index values.

Table 2
Descriptive statistics of cash holdings
The sample consists of firms included on the Compustat database in 1997. All data are for 1997. Cash hold is defined as cash and short-term investments scaled by assets. The Herfindahl-Hirschman Index is calculated individual manufacturing industries defined at the six-digit NAICS level (NAICS codes are from 31111-339999). Deviation from median industry capital-to-labor ratio is calculated by taking the absolute value of difference between a firm's ratio of net plant and equipment per employee and the median ratio in the firm's 6-digit SIC industry and then scaling this difference by the industry range of net plant and equipment per employee. Firm-industry stock return correlation is the regression coefficient on monthly stock returns of an industry equity weighted stock return index from a regression of monthly firm stock returns on monthly CRSP equally weighted stock returns and monthly industry index stock returns. There are 2,843 observations used to calculate the holdings statistics. p -values for whether means and medians differ from subsamples based on the Herfindahl-Hirschman Index, absolute deviation from median industry capital-to-labor ratio, and firm-industry stock return correlation are for the two-tailed difference in means test and the two-tailed Wilcoxon rank-sum test, respectively.

	Mean	Median
Panel A: Cash holdings for all sample firms		
All sample firms	0.216	0.105
Panel B: Cash holdings for Herfindahl-Hirschman Index-based subsamples		
Herfindahl-Hirschman Index < median subsample	0.198	0.087
Herfindahl-Hirschman Index > median subsample	0.235	0.130
p -value for differences between subsamples	0.000	0.000
Panel C: Cash holdings for absolute deviation from median industry capital-to-labor ratio-based subsamples		
Absolute deviation from median industry capital-to-labor ratio < median subsample	0.254	0.130
Absolute deviation from median industry capital-to-labor ratio > median subsample	0.179	0.084
p -value for differences between subsamples	0.000	0.000
Panel D: Cash holdings for firm-industry stock return correlation-based subsamples		

Table 2. Descriptive statistics of cash holdings.

Read:

- Table 1 documents large variation in industry concentration across manufacturing industries. Transportation equipment, beverage and tobacco, chemicals, and leather/allied products are among the most concentrated industries.
- Table 2 shows that cash holdings are larger in high-HHI industries, for firms closer to the industry technological core, and for firms with higher firm-industry stock return correlation.
- The univariate evidence already supports the paper's idea that firms exposed to more predation risk hold more internal liquidity.

Economic message: product-market structure and within-industry similarity are not merely background industry facts. They are correlated with balance-sheet liquidity.

Table 3

Descriptive statistics of currency swap usage

The sample consists of firms in the manufacturing sector that are part of the S&P 500 Index in 1997. All data are for 1997. The Herfindahl-Hirschman Index is calculated for individual manufacturing industries defined at the six-digit NAICS level (NAICS codes are from 311111 to 339999). Deviation from median industry capital-to-labor ratio is calculated by taking the absolute value of the difference between a firm's ratio of net plant and equipment per employee and the median ratio in the firm's four-digit SIC industry and then scaling this difference by the industry range of net plant and equipment per employee. Firm-industry stock return correlation is the regression coefficient on monthly stock returns of an industry equally weighted stock return index from regression of monthly firm stock returns on monthly CRSP equally weighted stock returns and monthly industry stock returns. There are 151 observations used to calculate the currency swap statistics. p -values for whether proportions differ between subsamples based on the Herfindahl-Hirschman Index, absolute deviation from median industry capital-to-labor ratio, and firm-industry stock return correlation are for the chi-squared test corrected for continuity.

	Proportion
Panel A: Swap usage for all sample firms	
All sample firms	0.808
Panel B: Swap usage for Herfindahl-Hirschman Index-based subsamples	
Herfindahl-Hirschman Index < median subsample	0.813
Herfindahl-Hirschman Index > median subsample	0.803
p -value for differences between subsamples	1.000
Panel C: Swap usage for absolute deviation from median industry capital-to-labor ratio-based subsamples	
Absolute deviation from median industry capital-to-labor ratio < median subsample	0.787
Absolute deviation from median industry capital-to-labor ratio > median subsample	0.829
p -value for differences between subsamples	0.651

Table 3. Descriptive statistics of currency swap usage.

Read:

- Table 3 shows that currency swap use is common among S&P 500 manufacturing firms, but simple univariate splits do not show strong differences across predation-risk groups.
- Table 4 provides multivariate evidence: both the HHI and the four-firm concentration ratio are positively associated with cash holdings.
- The estimated relation survives controls for foreign sales, diversification, size, market-to-book assets, R&D, capital expenditures, leverage, bond ratings, dividends, profitability, and industry operating-income volatility.

Economic message: once standard determinants of liquidity are held fixed, firms in more concentrated industries hold more cash, consistent with managing predation risk.

Table 4

Regressions of cash holdings on industry concentration measures

The sample consists of firms in the manufacturing sector that are included on Compustat during any part of 1993-2001 period. Models 1 and 3 are pooled OLS regression models in which significance levels for the estimates are different from zero using Newey and West (1987) corrected standard errors are in parentheses. Models 2 and 4 are OLS models that use time-series variable means in which significance levels for the estimates are different from zero corrected for heteroskedasticity using the White (1980) correction are in parentheses. The dependent variable is the natural logarithm of the sum of cash and short-term investments divided by book assets minus cash and short-term investments. The Industry Herfindahl-Hirschman Index four-firm concentration ratio are as of 1997 and are divided, respectively, by 1000 and ten for scaling purpose. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm's four-digit SIC code is used to calculate statistic if there are at least five firms in the firm's four-digit SIC code for all five of the years. If this criterion is met, the firm's three-digit SIC code is used. If this criterion is not met when using a firm's three-digit SIC code two-digit SIC code is used. Year dummies are included in Models 1 and 3. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2	3	4
Intercept	-1.986 (0.000)	-2.014 (0.000)	-2.214 (0.000)	-2.232 (0.000)
Industry Herfindahl-Hirschman index	0.224 (0.000)	0.174 (0.003)		
Industry four-firm concentration ratio			0.093 (0.000)	0.087 (0.000)
Foreign sales/total sales	0.844 (0.000)	1.077 (0.000)	0.814 (0.000)	1.048 (0.000)
Number of business segments	-0.142 (0.000)	-0.139 (0.000)	-0.139 (0.000)	-0.133 (0.000)
Natural logarithm of book value of assets	-0.038 (0.004)	-0.079 (0.001)	-0.037 (0.005)	-0.080 (0.001)
Market-to-book assets	0.151 (0.000)	0.211 (0.000)	0.151 (0.000)	0.210 (0.000)
R&D/sales	0.001 (0.002)	0.003 (0.047)	0.001 (0.002)	0.003 (0.049)
Capital expenditures/book assets	-3.018 (0.000)	-4.195 (0.000)	-2.986 (0.000)	-4.286 (0.000)
Total leverage	-0.165 (0.000)	-0.276 (0.000)	-0.165 (0.000)	-0.275 (0.000)
Bond rating dummy	-0.324 (0.000)	-0.165 (0.000)	-0.333 (0.000)	-0.184 (0.113)
Dividend dummy	-0.508 (0.000)	-0.490 (0.000)	-0.490 (0.000)	-0.470 (0.000)
Average operating income/book value of assets from year -3 to -1	-0.790 (0.000)	-0.998 (0.000)	-0.788 (0.000)	-0.995 (0.000)

Table 4. Cash holdings and industry concentration.

6.2 Hedging and intra-industry interdependence

Table 5

Probit models of the decision to use currency swaps on industry concentration measures

The sample consists of firms in the manufacturing sector that are part of the S&P 500 Index in 1997. The sample period is 1993–1997. The dependent variable is a dummy variable for whether the firm utilizes currency swaps during a particular year. The Industry Herfindahl-Hirschman Index and four-firm concentration ratio are as of 1997 and are divided, respectively, by 1000 and ten for scaling purposes. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm's four-digit SIC code is used to calculate this statistic if there are at least five firms in the firm's four-digit SIC code for all five of the years. If this criterion is not met the firm's three-digit SIC code is used. If this criterion is not met when using a firm's three-digit SIC code, the two-digit SIC code is used. Year dummies are included in the models. The pseudo- R^2 is computed as $1 - [\text{Log Likelihood of full model} / \text{Log Likelihood of a model that includes only the constant}]$. Marginal effects estimates are presented. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2
Intercept	−0.442 (0.004)	−0.426 (0.004)
Industry Herfindahl-Hirschman Index	0.069 (0.042)	
Industry four-firm concentration ratio		0.023 (0.016)
Foreign sales/total sales	0.722 (0.000)	0.761 (0.000)
Number of business segments	0.007 (0.615)	0.008 (0.597)
CEO fractional ownership	−0.421 (0.159)	−0.418 (0.163)
Natural logarithm of book value of assets	0.018 (0.386)	0.003 (0.865)
Market-to-book assets	0.002 (0.860)	0.001 (0.938)
R&D/sales	0.104 (0.604)	0.083 (0.681)
Capital expenditures/book assets	−0.297 (0.398)	−0.191 (0.582)
Total leverage	0.327 (0.008)	0.395 (0.001)
Bond rating dummy	0.133 (0.002)	0.131 (0.002)
Dividend dummy	−0.102 (0.017)	−0.098 (0.024)
Operating income/book value of assets	0.364 (0.115)	0.438 (0.055)
Coefficient of variation of median industry operating income/book value of assets	0.350 (0.275)	0.285 (0.310)
Pseudo- R^2	0.296	0.298
N	589	606

Table 5. Currency swap usage and industry concentration.

Table 6

Regressions of cash holdings on absolute value of capital-to-labor deviation and correlation of firm stock returns with industry stock returns

The sample consists of firms in the manufacturing sector that are included on Compustat during any part of the 1993–2001 period. Models 1 and 4 are pooled OLS regression models in which significance levels for whether estimates are different from zero using Newey and West (1987) corrected standard errors are in parentheses. Models 2 and 5 are OLS models that use time-series variable means in which significance levels for whether estimates are different from zero corrected for heteroskedasticity using the White (1980) correction are in parentheses. Model 3 is an OLS fixed effects model in which time-series variable means are subtracted from each variable in which significance levels for whether estimates are different from zero corrected for heteroskedasticity using the White (1980) correction are in parentheses. The dependent variable is the natural logarithm of the sum of cash and short-term investments divided by book assets minus cash and short-term investments. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm's four-digit SIC code is used to calculate this statistic if there are at least five firms in the firm's four-digit SIC code for all five of the years. If this criterion is not met the firm's three-digit SIC code is used. If this criterion is not met when using a firm's three-digit SIC code, the two-digit SIC code is used. Year and two-digit SIC dummies are included in the models. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2	3	4	5
Intercept	−2.450 (0.000)	−2.490 (0.000)	−0.101 (0.131)	−3.354 (0.000)	−3.222 (0.000)
Absolute value of capital-to-labor ratio	−0.342 (0.004)	−0.426 (0.064)	−0.265 (0.025)		
Correlation of firm stock returns with industry stock returns				0.511 (0.000)	0.419 (0.000)
Foreign sales/total sales	0.361 (0.000)	0.624 (0.000)	0.086 (0.228)	0.365 (0.000)	0.585 (0.000)
Number of business segments	−0.135 (0.000)	−0.126 (0.000)	−0.091 (0.000)	−0.109 (0.000)	−0.095 (0.000)
Natural logarithm of book value of assets	0.015 (0.278)	−0.027 (0.249)	0.102 (0.000)	0.039 (0.001)	−0.017 (0.430)
Market-to-book assets	0.111 (0.000)	0.154 (0.000)	0.037 (0.000)	0.114 (0.000)	0.155 (0.000)
R&D/sales	0.001 (0.001)	0.002 (0.059)	0.000 (0.021)	0.001 (0.021)	0.002 (0.157)
Capital expenditures/book assets	−1.942 (0.000)	−2.729 (0.000)	−1.914 (0.000)	−2.554 (0.000)	−3.116 (0.000)
Total leverage	−0.194 (0.001)	−0.202 (0.000)	−0.048 (0.000)	−0.125 (0.000)	−0.203 (0.000)
Bond rating dummy	−0.420 (0.000)	−0.352 (0.008)	−0.202 (0.002)	−0.406 (0.000)	−0.280 (0.015)
Dividend dummy	−0.393 (0.000)	−0.417 (0.000)	0.024 (0.691)	−0.123 (0.006)	−0.154 (0.054)
Average operating income/book value of assets from year −3 to −1	−0.669 (0.000)	−0.847 (0.000)	−0.192 (0.004)	−0.565 (0.004)	−0.781 (0.004)
Coefficient of variation of median industry operating income/book value of assets	0.197 (0.000)	0.629 (0.000)	−0.017 (0.593)	0.180 (0.000)	0.562 (0.000)
R^2 -adjusted	0.295	0.397	0.027	0.299	0.402
N	14,059	3,110	14,059	17,533	3,529

Table 6. Cash holdings, technological similarity, and return correlation.

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- Table 5 shows that firms are more likely to use currency swaps when their industries are more concentrated.
- Table 6 shifts from across-industry to within-industry evidence. Cash holdings are negatively related to the absolute capital-to-labor deviation measure, meaning firms closer to the technological core hold more cash.
- Cash holdings are also positively related to the correlation between firm stock returns and industry stock returns.
- The fixed-effect specification suggests that, when a firm's technology becomes more similar to its industry peers, it tends to increase cash holdings.

Economic message: predation risk is not simply an industry-level phenomenon. Within an industry, firms most similar to rivals and most exposed to common opportunity shocks appear to preserve more internal financing capacity.

Table 7
Probit models of the decision to use currency swaps on absolute value of capital-to-labor deviation and correlation of firm stock returns with industry stock returns

The sample consists of firms in the manufacturing sector that are part of the S&P 500 Index in 1997. The sample period is 1993–1997. The dependent variable is a dummy variable for whether the firm utilizes currency swaps during a particular year. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm’s four-digit SIC code is used to calculate this statistic if there are at least five firms in the firm’s four-digit SIC code for all five of the years. If this criterion is not met, the firm’s three-digit SIC code is used. If this criterion is not met when using a firm’s three-digit SIC code, the two-digit SIC code is used. Year and two-digit SIC dummies are included in the models. The pseudo- R^2 is computed as $1 - [\text{Log Likelihood of full model} / \text{Log Likelihood of a model that includes only the constant}]$. Marginal effects estimates are presented. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2
Intercept	−0.125 (0.377)	0.188 (0.999)
Absolute value of capital-to-labor ratio deviation	−0.252 (0.002)	
Correlation of firm stock returns with industry stock returns		0.206 (0.000)
Foreign sales/total sales	0.439 (0.000)	0.492 (0.000)
Number of business segments	0.019 (0.168)	0.040 (0.004)
CEO fractional ownership	−0.042 (0.866)	−0.210 (0.335)
Natural logarithm of book value of assets	−0.007 (0.676)	0.008 (0.587)
Market-to-book assets	−0.004 (0.672)	0.002 (0.829)
R&D/sales	0.013 (0.944)	−0.094 (0.575)
Capital expenditures/book assets	0.025 (0.938)	−0.244 (0.388)
Total leverage	0.388 (0.001)	0.390 (0.000)
Bond rating dummy	0.139 (0.001)	0.075 (0.040)
Dividend dummy	−0.169 (0.000)	−0.031 (0.344)
Operating income/book value of assets	0.619 (0.005)	0.594 (0.001)
Coefficient of variation of median industry operating income/book value of assets	0.346 (0.301)	0.354 (0.173)
Pseudo- R^2	0.387	0.415
N	488	589

Table 7. Currency swap usage, technological similarity, and return correlation.

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- Table 7 mirrors Table 6 for currency swaps. Firms closer to the industry technological core are more likely to use currency swaps.
- Firms whose stock returns co-move more with industry stock returns are also more likely to use currency swaps.
- Table 8 tests whether cash holdings respond more strongly to predation-risk proxies in high-growth-opportunity industries.
- Interactions with high industry market-to-book are generally significant in the predicted direction for cash holdings.

Economic message: the relation between predation-risk proxies and cash is stronger when there are more investment opportunities at stake, which supports the strategic-financing interpretation rather than a purely mechanical industry effect.

Table 8

Regression models examining the effects of industry growth opportunities

The sample consists of firms in the manufacturing sector that are included on Compustat during any part of the 1993–2001 period. Models 1–4 are pooled OLS regression models in which significance levels for whether estimates are different from zero using Newey and West (1987) corrected standard errors are in parentheses. TI dependent variable is the natural logarithm of the sum of cash and short-term investments divided by book assets minus cash and short-term investments. The Industry Herfindahl-Hirschman Index and four-firm concentration ratio are as of 1997 and are divided, respectively, by 1000 and ten for scaling purposes. The high industry market-to-book dummy takes on a value of one if a firm’s median four-digit SIC industry market-to-book ratio is above the sample median for the year and takes on a value of zero otherwise. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm’s four-digit SIC code is used to calculate this statistic if there are at least five firms in the firm’s four-digit SIC code for all five of the years. If this criterion is not met the firm’s three-digit SIC code is used. If this criterion is not met when using a firm’s three-digit SIC code, the two-digit SIC code is used. Year dummies are included in models 1 and 2. Year and two-digit SIC dummies are included in models 3 and 4. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2	3	4
Intercept	−2.762 (0.000)	−2.845 (0.000)	−2.757 (0.000)	−2.876 (0.000)
Industry Herfindahl-Hirschman Index	0.028 (0.502)			
Industry Herfindahl-Hirschman Index • high industry market-to-book dummy	0.362 (0.000)		0.040 (0.000)	
Industry four-firm concentration ratio			0.101 (0.000)	
Industry four-firm concentration ratio • high industry market-to-book dummy				0.012 (0.932)
Absolute value of capital-to-labor ratio deviation				−0.646 (0.007)
Absolute value of capital-to-labor ratio deviation • high industry market-to-book dummy				
Correlation of firm stock returns with industry stock returns				0.155 (0.001)
Correlation of firm stock returns with industry stock returns • high industry market-to-book dummy				0.610 (0.000)
High industry market-to-book dummy	0.939 (0.000)	0.750 (0.000)	0.885 (0.000)	0.163 (0.011)
Market model residual variance	0.389 (0.127)	0.324 (0.203)	0.610 (0.027)	−0.769 (0.011)
Foreign sales/total sales	0.560 (0.000)	0.553 (0.000)	0.319 (0.000)	0.395 (0.000)
Number of business segments	−0.071 (0.000)	−0.071 (0.000)	−0.104 (0.000)	−0.081 (0.000)
Natural logarithm of book value of assets	−0.018 (0.149)	−0.022 (0.068)	0.013 (0.342)	0.014 (0.253)
Market-to-book assets	0.122 (0.000)	0.122 (0.000)	0.104 (0.000)	0.106 (0.000)
R&D/sales	0.001 (0.003)	0.001 (0.003)	0.001 (0.001)	0.000 (0.091)
Capital expenditures/book assets	−2.868	−2.905	−2.160	−2.673

Table 8. Industry growth opportunities, part 1.

6.3 Investment behavior and product-market strategy

Table 8 (continued)

Model	1	2	3	4
Bond rating dummy	-0.412 (0.000)	-0.414 (0.000)	-0.448 (0.000)	-0.408 (0.000)
Dividend dummy	-0.212 (0.000)	-0.207 (0.000)	-0.218 (0.000)	-0.084 (0.054)
Average operating income/book value of assets from year -3 to -1	-0.630 (0.000)	-0.624 (0.000)	-0.567 (0.000)	-0.492 (0.000)
Coefficient of variation of median industry operating income/book value of assets	0.074 (0.099)	0.076 (0.084)	0.092 (0.056)	0.088 (0.052)

Table 8. Industry growth opportunities, continued.

Read:

- The continuation of Table 8 reports additional controls and confirms that the growth-opportunity interaction results are not driven only by standard cash-holdings determinants.
- Table 9 changes the dependent variable from financial policy to real investment behavior.
- The dependent variable equals one when a firm increases capital expenditures plus R&D while industry investment is falling.
- High cash holdings strongly increase the probability that a firm increases investment.

Economic message: cash is valuable because it gives firms the option to invest when others cannot. This is the key bridge from financial policy to real strategic behavior.

Table 9

Probit models of the decision to increase annual research and development and capital expenditures

The sample consists of firms in the manufacturing sector that are included on Compustat during any part of the 1993-2001 period. The dependent variable equals one if from year_{t-1} to year_t, capital expenditures and research and development expenses scaled by book assets increases and takes on a value of zero if this ratio decreases between these 2 years. One-year change in industry-level investment is the change in the median four-digit SIC industry ratio of capital expenditures and research and development expenses scaled by book assets from year_{t-1} to year_t. Absolute change in industry-level investment during year when industry-level investment decreases is the absolute value of the 1-year change in the median four-digit SIC industry ratio of capital expenditures and research and development expenses scaled by book assets from year_{t-1} to year_t during years when this ratio decreases. High cash holdings takes on a value of one if the natural logarithm of the sum of cash and short-term investments divided by book assets minus cash and short-term investments is above the sample period median value and takes a value of zero otherwise. The high Herfindahl-Hirschman Index, four-firm concentration ratio, and high correlation of firm stock returns with industry stock returns dummies take on a value of one if the firm value for one of these variables is above the sample median and takes on a value of zero otherwise. The low absolute value of capital-to-labor deviation dummy takes on a value of one if the firm value for this variable is below the sample median and takes on a value of zero otherwise. One-year change in industry sales is the change in industry sales defined by a firm's four-digit SIC industry from year_{t-1} to year_t. Industry-adjusted total leverage is a firm's total leverage minus the median value for total leverage that year in its four-digit SIC industry. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm's four-digit SIC code is used to calculate this statistic if there are at least five firms in the firm's four-digit SIC code for all five of the years. If this criterion is not met the firm's three-digit SIC code is used. If this criterion is not met when using a firm's three-digit SIC code, the two-digit SIC code is used. Two-digit SIC dummies are included in the models. The pseudo-R² is computed as 1 - [Log Likelihood of full model/Log Likelihood of a model that includes only the constant]. Marginal effects estimates are presented. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2	3	4	5	6
Intercept	-0.030 (0.228)	-0.067 (0.055)	-0.099 (0.006)	-0.096 (0.008)	-0.105 (0.013)	-0.107 (0.003)
One-year change in industry-level investment	2.051 (0.000)					
Absolute change in industry-level investment during years when industry-level investment decreases		-0.964 (0.000)	-1.719 (0.000)	-1.904 (0.000)	-1.661 (0.000)	-1.040 (0.001)
High cash holdings dummy			0.081 (0.000)	0.080 (0.000)	0.095 (0.000)	0.092 (0.000)
High industry Herfindahl-Hirschman Index dummy			0.000 (0.993)			
High industry four-firm concentration ratio dummy				-0.005 (0.672)		
Low absolute value of capital-to-labor ratio deviation dummy					0.000 (0.978)	
High correlation of firm stock returns with industry stock returns dummy						-0.005 (0.675)
Absolute change in industry-level investment during year when industry-level investment decreases + high cash holdings dummy + high industry Herfindahl-Hirschman Index dummy			0.836 (0.041)			
Absolute change in industry-level investment during year when industry-level investment decreases + high cash holdings dummy + high industry four-firm concentration ratio dummy				1.044 (0.010)		

Table 9. Decision to increase investment, part 1.

Table 9 (continued)

Model	1	2	3	4	5	6
Absolute change in industry-level investment during year when industry-level investment decreases * high cash holdings dummy * low absolute value of capital-to-labor ratio deviation dummy					0.877 (0.035)	
Absolute change in industry-level investment during year when industry-level investment decreases * high cash holdings dummy * high correlation of firm stock returns with industry stock returns dummy						-0.088 (0.807)
One-year change in industry sales	0.002 (0.770)	-0.005 (0.559)	-0.006 (0.548)	-0.006 (0.526)	-0.008 (0.408)	-0.006 (0.546)
Natural logarithm of book value of assets	0.003 (0.177)	0.003 (0.369)	0.004 (0.205)	0.004 (0.195)	0.006 (0.141)	0.005 (0.177)
Industry-adjusted total leverage	-0.069 (0.000)	-0.047 (0.000)	-0.032 (0.004)	-0.033 (0.003)	-0.018 (0.166)	-0.033 (0.003)
Bond rating dummy	-0.027 (0.107)	-0.011 (0.603)	-0.000 (0.988)	-0.004 (0.859)	-0.011 (0.661)	-0.005 (0.835)
Dividend dummy	-0.048 (0.000)	-0.051 (0.000)	-0.047 (0.001)	-0.046 (0.001)	-0.045 (0.005)	-0.048 (0.001)
Operating income/book value of assets	0.066 (0.000)	0.072 (0.000)	0.104 (0.000)	0.104 (0.000)	0.102 (0.000)	0.102 (0.000)
Coefficient of variation of median industry operating income/book value of assets	0.011 (0.679)	-0.012 (0.710)	-0.014 (0.665)	-0.009 (0.777)	0.002 (0.944)	0.016 (0.616)
Pseudo- R^2	0.023	0.013	0.018	0.018	0.019	0.018
N	18,407	9,371	9,061	9,232	7,024	9,234

Table 9. Decision to increase investment, continued.

Read:

- The continued part of Table 9 shows that the interaction of industry investment declines, high cash holdings, and predation-risk proxies is positive and significant for the HHI, four-firm concentration, and technological-core measures.
- The stock-return-correlation interaction is not significant, which suggests that not all interdependence proxies work equally well in the investment-behavior test.
- Table 10 shows that firms with higher industry-adjusted cash holdings are less likely to use currency swaps.
- Currency swap usage is more likely when cash holdings vary more within the industry, consistent with firms using derivatives when cash positions differ across rivals.

Economic message: the investment tests support the strategic-benefit story: cash-rich firms facing high predation risk are able to invest against the industry cycle. The derivatives tests suggest that cash and hedging can substitute for each other as ways to protect investment capacity.

7 Key mechanisms and interpretation

7.1 Why product-market interdependence matters

Predation risk is stronger when a firm's investment opportunities overlap with rivals. In that case, a rival's investment can directly reduce the firm's future market share or strategic position. Industry

Table 10

Probit models examining the association between cash holdings and the decision to use currency swaps

The sample consists of firms in the manufacturing sector that are part of the S&P 500 Index in 1997. The sample period is 1993–1997. The dependent variable is a dummy variable for whether the firm utilizes currency swaps during a particular year. Industry-adjusted cash holdings is measured as cash holdings/book assets for a firm minus the median ratio in the firm's four-digit SIC industry. Standard deviation of industry cash holdings is measured as the standard deviation of cash holdings/book assets during a given year in a firm's four-digit SIC industry. Coefficient of variation of median industry operating income/book value of assets is calculated as the absolute value of the coefficient of variation of median industry operating income/book value of assets during the 5 years prior to a particular firm year and is then scaled by 1000. A firm's four-digit SIC code is used to calculate this statistic if there are at least five firms in the firm's four-digit SIC code for all five of the years. If this criterion is not met the firm's three-digit SIC code is used. If this criterion is not met when using a firm's three-digit SIC code, the two-digit SIC code is used. Year dummies are included in the models. The Pseudo- R^2 is computed as $1 - [\text{Log Likelihood of full model} / \text{Log Likelihood of a model that includes only the constant}]$. Marginal effects estimates are presented. Significance levels for whether estimates are different from zero are in parentheses.

Model	1	2
Intercept	-0.352 (0.018)	-0.501 (0.001)
Industry-adjusted cash holdings	-0.292 (0.009)	
Standard deviation of industry cash holdings		0.385 (0.090)
Foreign sales/total sales	0.696 (0.000)	0.672 (0.000)
Number of business segments	0.004 (0.765)	0.008 (0.596)
CEO fractional ownership	-0.221 (0.470)	-0.260 (0.398)
Natural logarithm of book value of assets	0.007 (0.703)	0.014 (0.462)
Market-to-book assets	0.003 (0.827)	-0.001 (0.952)
R&D/sales	0.053 (0.794)	0.074 (0.717)
Capital expenditures/book assets	-0.236 (0.493)	0.002 (0.995)
Total leverage	0.432 (0.000)	0.467 (0.000)
Bond rating dummy	0.127 (0.003)	0.123 (0.003)
Dividend dummy	-0.127 (0.004)	-0.098 (0.022)
Operating income/book value of assets	0.482 (0.034)	0.484 (0.034)
Coefficient of variation of median industry operating income/book value of assets	0.112 (0.651)	0.125 (0.611)
Pseudo- R^2	0.300	0.296
N	604	601

Table 10. Cash holdings and the decision to use currency swaps.

concentration, technological similarity, and firm-industry return correlation are all attempts to capture this interdependence.

7.2 Why cash holdings are strategic

Cash holdings let a firm finance planned or unplanned investments when external finance is costly or when industry cash flows deteriorate. In the product-market setting, cash can also be offensive: a cash-rich firm can increase investment when rivals are cutting back.

7.3 Why hedging is strategic

Currency swaps reduce exposure to cash-flow shocks and therefore help firms maintain investment capacity. The paper interprets derivatives as one way to protect against underinvestment when product-market conditions deteriorate.

7.4 Cash and derivatives as substitutes

If cash and derivatives both help preserve investment capacity, firms may not need both. The negative relation between industry-adjusted cash holdings and currency swap usage supports this substitution view.

8 Limitations

- **Predation risk is proxied, not observed directly.** Industry concentration, technological similarity, and stock-return correlation can capture other economic forces besides predation risk.
- **Derivatives evidence is limited to S&P 500 manufacturing firms and currency swaps.** The results may not generalize to smaller firms, nonmanufacturing firms, or other hedging instruments.
- **Causality is difficult.** Firms choose cash, hedging, and product-market strategy jointly. The evidence is consistent with strategic risk management but is not a randomized design.
- **Stock-return correlation is potentially endogenous.** Cash holdings and hedging can themselves affect stock-return behavior, so causality is less clean for that proxy.
- **The welfare effect is not directly measured.** The paper shows that firms manage predation risk and sometimes invest against industry downturns, but it does not prove whether this behavior increases total welfare or only redistributes market share.

9 Short summary

Haushalter, Klasa, and Maxwell examine whether firms use cash holdings and derivatives to manage predation risk arising from product-market competition. They argue that predation risk is higher when a firm's investment opportunities are more interdependent with rivals. Using manufacturing firms in the 1990s, they show that cash holdings and currency swap usage are higher when industries are more concentrated, when firms are closer to the industry technological core, and when firm stock

returns co-move more with industry stock returns. The cash relation is stronger in high-growth-opportunity industries. They also show that cash-rich firms with high predation risk are more likely to increase investment when industry investment declines. Finally, they find evidence that cash holdings and currency swaps are substitutes.

10 Conclusion

The paper's main lesson is that corporate liquidity and hedging policies cannot be understood only as responses to taxes, bankruptcy costs, or generic financing constraints. They are also shaped by the firm's product-market environment. Firms appear to preserve internal financing capacity when rivals could exploit their inability to invest. In this sense, cash holdings and derivatives are strategic tools that link financing policy, investment policy, and competitive behavior.